

FL Simulation Results

File: test_A6

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 85.0% (0.85)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 85.0% (0.85) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8747
Clean + DP Protection Accuracy: 0.8736
Poisoned Accuracy: 0.7872
Poisoned + DP Protection Accuracy: 0.7872
Drop (Clean - Poisoned): 0.0875
Drop (Clean - Init): 0.0819
Drop (Clean DP - Init): 0.0808
Drop (Poisoned - Init): -0.0056
Drop (Poisoned DP - Init): -0.0056
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8804
Clean Recall: 0.8747
Clean F1 Score: 0.8740

Clean DP Precision: 0.8795
Clean DP Recall: 0.8736
Clean DP F1 Score: 0.8728

Poisoned Precision: 0.7997
Poisoned Recall: 0.7872
Poisoned F1 Score: 0.7863

DP Protection Precision: 0.7994
DP Protection Recall: 0.7872
DP Protection F1 Score: 0.7866

Summary

FL Simulation Complete
Task: 2f371002-c056-40cf-90a8-c72219342ad9
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8747
Clean DP Accuracy: 0.8736
Poisoned Accuracy: 0.7872
Data Poison Protection Accuracy: 0.7872
Drop (Clean - Poisoned): 0.0875
Drop (Clean - Init): 0.0819
Drop (Clean DP - Init): 0.0808
Drop (Poisoned - Init): -0.0056
Drop (Poisoned_DP - Init): -0.0056
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8804
Clean Recall: 0.8747
Clean F1 Score: 0.8740
Clean DP Precision: 0.8795
Clean DP Recall: 0.8736
Clean DP F1 Score: 0.8728
Poisoned Precision: 0.7997
Poisoned Recall: 0.7872
Poisoned F1 Score: 0.7863
DP Protection Precision: 0.7994
DP Protection Recall: 0.7872
DP Protection F1 Score: 0.7866